

Readaptation of the SING Algorithm to the Italian Case-Study Using Open Data

1 Introduction

In this work, we aim to generate a synthetic electrical grid within Italy using open data sources, focusing specifically on medium voltage (MV) networks linked to high voltage (HV) substations. This project is designed as a RESTful API to handle geospatial data processing for electric grid infrastructure. The main components are responsible for reading georeferenced input data, and outputting grid information in a GeoJSON format. The API is set up to handle optional data inputs, in order to use them in case they are available, and to use open data in case they are missing.

Leveraging the dataset provided by Gestore dei Servizi Energetici [1], we can delineate the geographical areas in which HV to MV substations are active. The methodology used here is based on the SING software framework [2], which enables the generation of synthetic grids by integrating data from publicly available datasets.

The essential data inputs for this synthetic grid model include:

- A defined region of interest for the grid generation process;
- Precise locations of primary substations within this region;
- Spatial data on building locations;
- Road network geometries;
- Load profiles for buildings;
- Characteristics of both high voltage and medium voltage grid structures.

A significant portion of these data requirements is met using OpenStreetMap [3], though data completeness and accuracy are variable across regions. To validate OSM's primary substation data, a comparative analysis with the GSE dataset is performed, ensuring the accuracy and reliability of the input data. An exploratory analysis of this dataset then provides insight into its applicability for synthetic grid generation.

The characteristics of Italy's HV and MV grids are derived from TIDE [4] and Codice di Rete [5], enabling the adaptation of SING's algorithms to Italian network conditions. For building load profiles, the SING framework typically employs ResStock [6] and ComStock [7] profiles,

modeled using EnergyPlus software [8]. While ResStock represents residential energy demand, ComStock simulates commercial building consumption patterns, with quarter-hourly profiles that span across eight load categories. However, as Italian load profiles may differ from these U.S.-based models, we reference the Italian regulatory authority, ARERA [9], which provides average consumption data by contract power and NUTS3 region. Additionally, GSE offers hourly standard load profiles [1], which can be matched with ARERA data to create localized hourly profiles. For commercial load estimates, Terna's statistical reports [10] serve as a source for yearly consumption data, which can be applied to commercial GSE profiles to achieve a comparable hourly breakdown.

This framework leverages a comprehensive set of open data resources to create a realistic synthetic grid that aligns with the unique characteristics of the Italian power system, enabling robust analysis and optimization in line with local energy demands and infrastructure.

2 Methodology

The region of interest which is considered in this work is the border within which a primary substation transfers energy to final users. Such borders are provided by Gestore dei Servizi Energetici (GSE) [1].

2.1 Primary Substations Data Validation

Primary substations' georeferenced data can be fetched from Open Street Map (OSM) via Overpass [3]. Substation data is retrieved from the Overpass API within the boundaries of a given polygon or multipolygon. The geometry type of the input is checked, its bounds are extracted, and a query string for the Overpass API is written. The response from the API, containing the substation data, is returned.

The metadata of substations in Italy are analyzed in order to reconcile discrepancies between substation data from Italy's official energy service manager, GSE, and OpenStreetMap (OSM). Challenges include multiple substations per border in OSM data and operator inconsistencies. This part of the work, then, relies on two data sources:

- **GSE Data:** The official data is sourced from Gestore dei Servizi Energetici SpA (GSE, Italy). This dataset includes information about Medium Voltage (MV) substation boundaries, AC codes, operator names, and a total of 2107 records.
- **OpenStreetMap (OSM) Data:** The OSM data includes tags for power substations, with a total of 34158 records for Italy. The data includes various attributes of the substations such as their status (active, disused, abandoned, etc.), type (distribution, transmission, etc.), operator, and voltage level.

The methodology for this research involves several steps:

1. **Data Extraction:** Extract all substation records from both GSE and OSM data sources.
2. **Data Preparation:** Create sets of all the tag values for substations in Italy from both datasets. This includes filtering out all substations that are ‘disused’, ‘abandoned’, etc., and keeping those with the tag `substation = "distribution"`.
3. **Data Comparison:** Compare the data from both sources. This involves joining the data and checking for inconsistencies, such as borders with no substations or borders with more than one substation.
4. **Operator Filtering:** Filter the data based on operators. Join the operator data with the official ones from the GSE data.
5. **Data Correction:** For borders with more than one substation, store the OSM substations in a dedicated dataset. Filter these substations by operator and join with the official operators. If the join finds nothing, keep the data. If the join finds something, keep only the correct substation.

This methodology allows for a systematic comparison and reconciliation of the two datasets, leading to a more accurate and consistent dataset for substations in Italy. All the metadata labels found in OSM are reported in the Supplementary Materials.

We construct a query string for the Overpass API. The query is designed to search for ways with the tag `"power"="substation"` within a specified area. The area is defined by the `state` parameter, which in our case takes the value “Italy”. The query is written in code block (1).

(1)

```
def analyze_substations_tags_values_by_state_distribution(
    self,
    state: str,
    property_names: List[str]
):
    query = f"""
    area[name="{state}"]->.searchArea;
    (
        way["power"="substation"](area.searchArea);
    ); out geom;
    """
    response = self._api.Get(query)
```

We focus on substations that have the metadata label “substation” with value “distribution”, since those should correspond to HV/MV primary substations, which may correspond to the borders provided by GSE. Moreover, the substations with metadata values such as “abandoned”, “disused”, etc. The remaining properties are listed in the Supplementary materials (attached Excel file, `Italia_substation_analysis (3)`).

For each primary substation border, a cross check is performed with the open street map primary substations. In particular, if one of the substation has the same operator as the one reported by GSE, such substation is selected from the subset. If no corresponding operators

are found, only the substation nearest to the centroid of the border is kept. If no substations are found in the border, the centroid is chosen as primary substation location.

2.2 Synthetic Grid Generation

The **Synthetic Infrastructure Network Generation (SING)** framework is designed to create representative power distribution system models that are realistic, geographically accurate, and tailored to specific operational requirements. SING leverages real-world data sources and advanced modeling techniques to simulate distribution grid behavior, facilitating the testing of grid performance under varied conditions and helping researchers develop optimized strategies for grid management. Here's a step-by-step methodology of the SING framework, structured around key processes such as data acquisition, network topology generation, load profile development, and validation.

Data Acquisition and Geographic Mapping

The foundation of SING's synthetic grid generation relies on integrating geographic and infrastructure data from reliable sources, primarily OpenStreetMap (OSM) for street layouts and the GSE for local grid data. Schneider et al. (2017) highlight the importance of ensuring that synthetic models closely mimic real-world infrastructure to enable effective testing and planning for grid expansions and optimizations [11]. Geographic mapping from these sources allows SING to generate a grid model that mirrors the physical arrangement of distribution feeders and substations in the actual environment.

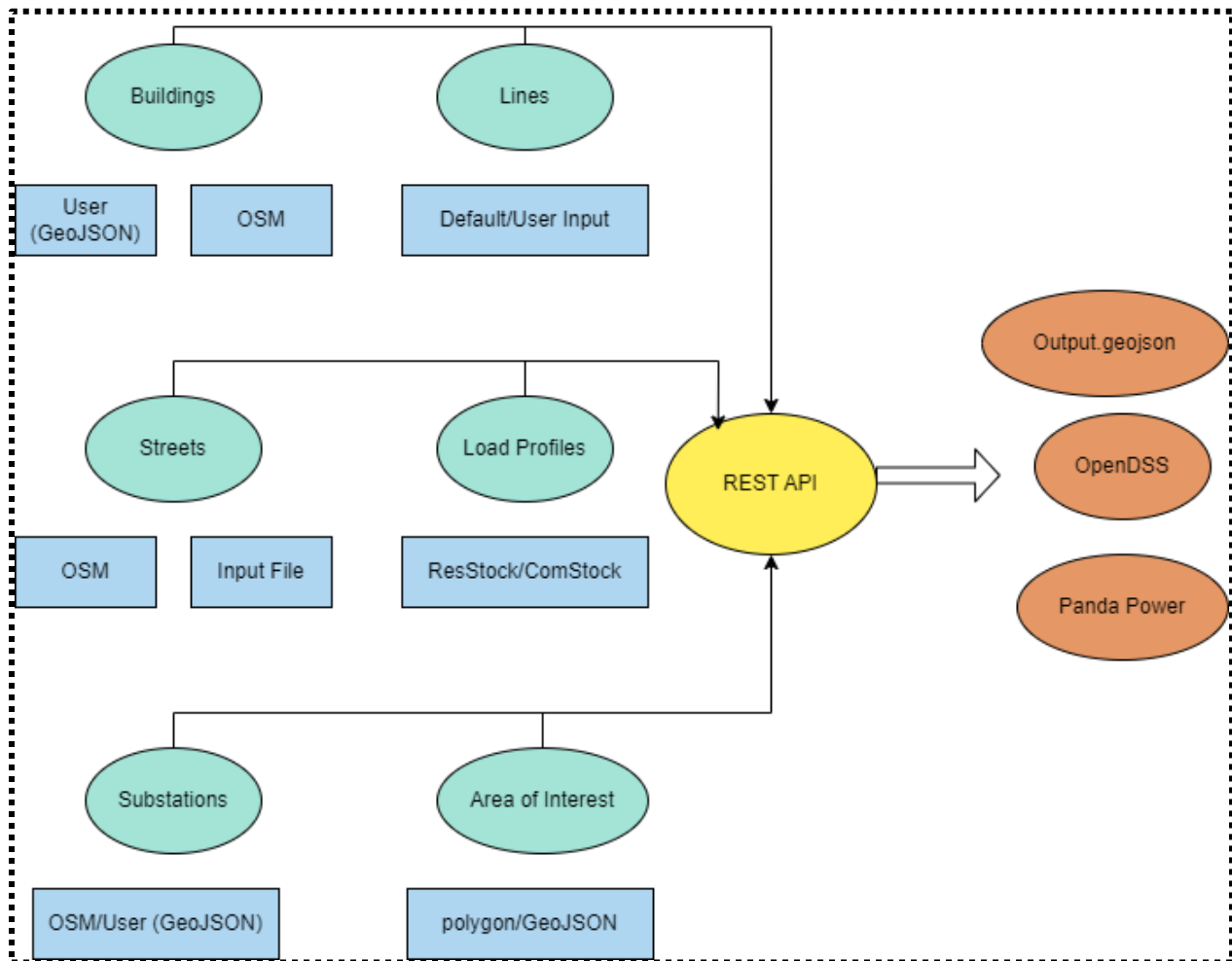
The data acquisition process within SING follows these steps:

1. **Extraction of Street and Substation Data:** SING accesses OSM to obtain street layout and substation positions, forming the basis for realistic grid topologies.
2. **Boundary Identification:** Using the geographic data, boundaries of regions and service areas are defined to specify the operational limits for each synthetic grid model.
3. **Node Placement:** With substation and feeder locations set, synthetic nodes representing consumers and distribution points are plotted along street maps for accurate load distribution simulations.

Load Profile and Consumption Pattern Generation

Creating realistic load profiles is central to SING's goal of accurately replicating grid behavior. As per Postigo Marcos et al. (2017), incorporating a wide range of consumption patterns enables synthetic grids to handle diverse operational scenarios, from peak demand to low consumption periods [14]. The load profile generation process involves:

1. **Categorization of Consumer Types:** Nodes are categorized as residential, commercial, or industrial, each with distinct consumption patterns based on typical load profiles observed in real distribution networks.
2. **Hourly Load Variability Simulation:** For each consumer type, load variations are programmed to mimic daily and seasonal demand fluctuations. The profiles are drawn from historical data and adjusted for typical usage patterns of each category.



SING Flowchart

This flowchart represents the workflow of the SING project, detailing how various input data sources are processed through a REST API to generate specific outputs. The chart is structured with inputs on the left, the REST API in the center, and the outputs on the right.

Inputs:

1. **Buildings:**
 - **User (GeoJSON):** User-provided geospatial data in GeoJSON format.
 - **OSM (OpenStreetMap):** Data extracted from OpenStreetMap.

2. **Streets:**
 - **OSM (OpenStreetMap):** Street data from OpenStreetMap.
 - **File:** Street data provided by the user in a file.
3. **Substations:**
 - **OSM (OpenStreetMap):** Substation data from OpenStreetMap.
 - **User (GeoJSON):** User-provided substation data in GeoJSON format.
4. **Lines:**
 - **Default from OSM:** Default line data from OpenStreetMap.
 - **User Input:** Line data provided by the user.
5. **Load Profiles:**
 - **ResStock:** Residential building load profiles.
 - **ComStock:** Commercial building load profiles.
6. **Area of Interest:**
 - **Polygon:** The geographic boundary defined as a polygon.
 - **GeoJSON:** Area of interest provided in GeoJSON format.

REST API: The central component of the workflow is the REST API, which processes the input data. It integrates data from various sources to create a cohesive and realistic synthetic electrical grid model.

Outputs:

1. **Output.geojson:** A GeoJSON file that visualizes the generated synthetic grid, including nodes, lines, and substations.
2. **OpenDSS:** Output files compatible with OpenDSS (Open Distribution System Simulator) for further power flow analysis.
3. **Panda Power:** Output files compatible with Panda Power, a tool for simulating and analyzing electrical power networks.

Lines: Refer to electrical conductors connecting components of the power distribution network, crucial for transferring electricity from substations to consumers. They have specific impedance and capacity values to mimic real-world electrical behavior, are routed realistically using OSM data, and are designed in a radial network topology for proper load distribution and fault isolation.

Load Profiles refer to patterns of electrical consumption over time for different buildings and consumers. These are crucial for simulating and understanding grid demand.

- **ResStock:** Provides residential energy use patterns.
- **ComStock:** Provides commercial energy use patterns.
- **ARERA, GSE, Terna:** Offer additional data to tailor load profiles to local conditions in Italy.

These profiles help in grid planning, optimization, and integrating renewable energy by predicting energy needs accurately.

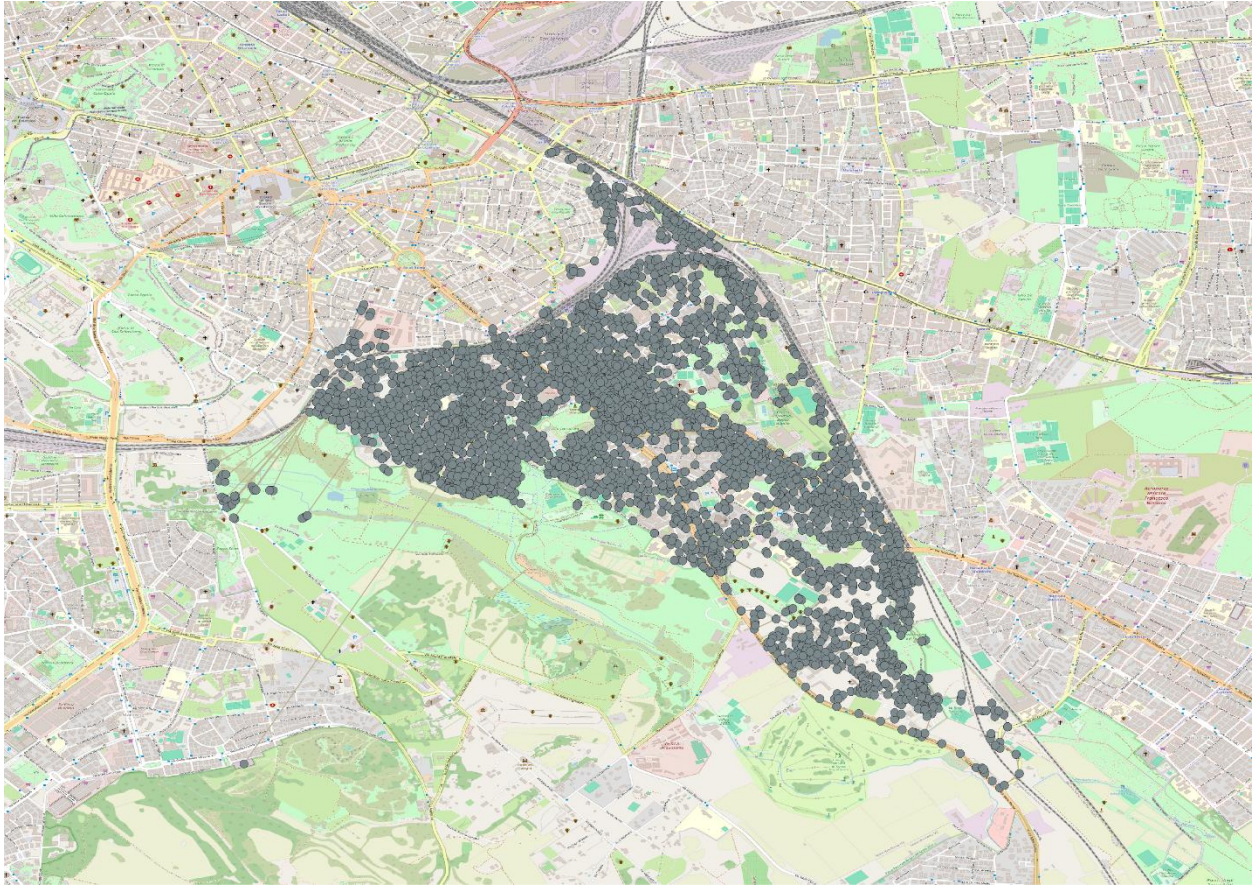
Substations refer to facilities in the electrical distribution network where voltage is transformed from high to medium voltage and managed for distribution to end-users. They are crucial nodes in the power grid, ensuring efficient and reliable electricity delivery.

Output GeoJson:

The output GeoJSON file represents the results of the synthetic grid generation process within the Italian context, based on the SING framework and open data sources. It visualizes key geospatial features related to the network, including:

1. **Primary Substation Locations:** The file highlights the precise geographic positions of substations, ensuring that the network's layout is accurately modeled within the defined region of interest.
2. **Network Connectivity:** It includes the distribution grid connections, linking substations and other relevant points, such as buildings and streets, following the radial network topology used in the SING algorithm.
3. **Feeder and Node Data:** Feeder lines, which represent the distribution paths from substations to the end users, are marked, with nodes indicating areas of load consumption such as residential, commercial, and industrial buildings.
4. **Grid Characteristics:** The file also integrates essential grid data, such as line impedance and capacity values, ensuring that the synthetic grid replicates real-world electrical characteristics.

The visualization encapsulated in the GeoJSON file allows for spatial analysis of the synthetic grid, facilitating further simulations of load flow, fault conditions, and grid optimization scenarios. This file serves as an essential tool for analyzing and validating the synthetic distribution network generated from the available data.



3 Results

3.1 Primary Substations Data Validation

The total number of official borders is 2107. Out of these, 1500 borders contain exactly one substation, and 348 borders contain more than one substation. Therefore, the number of borders with no substations at all is 259.

Step	Description	Result
Data Extraction and Preparation	Reading the official file containing substation borders according to GSE	2107 official borders
	Initial filtering on the OSM data	2336 substations from OSM

Step	Description	Result
Spatial Join Operation	Performing a spatial join operation between the substation and borders GeoDataFrames	488 borders without an OSM substation inside
Data Grouping	Grouping the data after initial filtering on the OSM data	1500 borders contain only one substation from OSM, 348 borders have more than one OSM substation
Data Visualization	Creating a scatter plot map showing the geographical distribution of single and multiple substations within their respective borders	Scatter plot map created
Operator Filtering	Defining a dictionary mapping the names of operators to a list of alternative names, and filtering the data based on the operator of each substation	Dictionary called operator_map defined
Final Output Information	Considering the operator of each substation	413 substations located within borders that contain more than one substation, 159 substations located within borders that contain exactly one substation

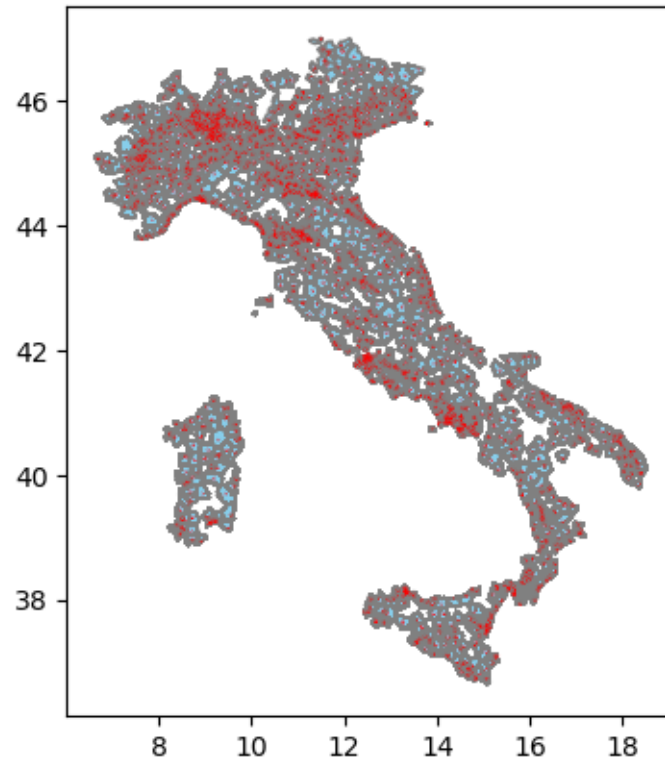


Figure 1: *single_substation_nodes_gdf*

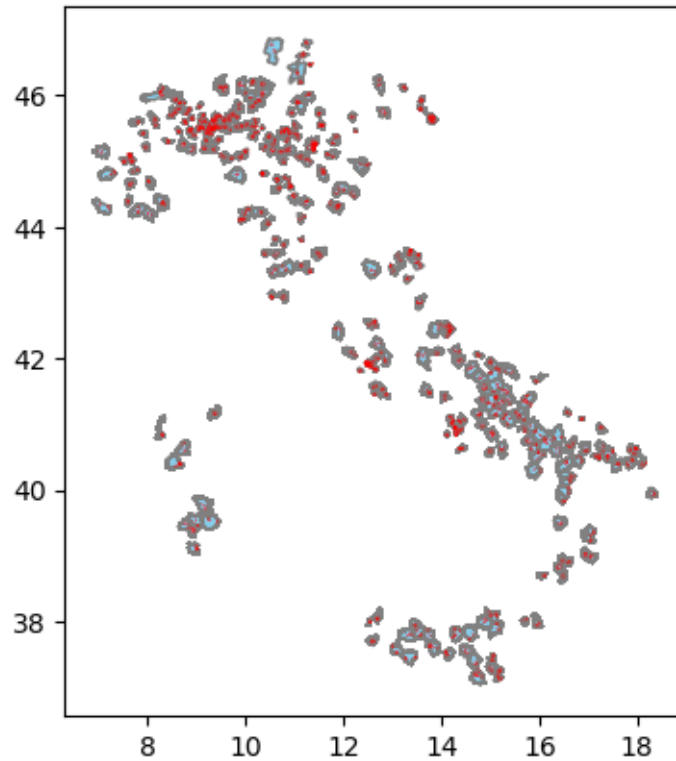


Figure 2: *multiple_substation_nodes_gdf (more than 1 substations)*

4 References

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